

✓ 1. Give an Exact Word Ceiling (Most Effective)

Principle

Be **explicit and measurable** when defining word limits.

Example

Explain machine learning in **under 50 words**.

Why This Works

The model attempts to optimize:

- Instruction compliance
- Pattern matching with training data

Clear numeric constraints significantly improve accuracy.

✓ 2. Specify “Maximum” — Not “Around”

Principle

Precision matters. Vague wording produces unpredictable results.

Strong Instruction

Maximum **40 words**.

Weak Instruction

Keep it short.

Key Insight

Ambiguous phrasing increases variability in output length.

3. Combine Word Limits with Structure (Highly Reliable)

Principle

Adding structure dramatically improves compliance.

Example

Answer in **3 bullet points**, each under **10 words**.

Why This Works

This approach constrains:

- Total word count
- Output format
- Token distribution

Structured constraints reduce randomness.

Why Word Limits Are Not Perfect

Core Limitation

LLMs internally count **tokens**, not words.

They estimate word length rather than measuring it precisely.

Result

Outputs may vary by approximately **±10%** from the requested word limit.

This behavior is normal.

If You Need Strict Control (Production Systems)

Use the `max_tokens` Parameter

Tokens are the **only enforceable output unit**.

Conversion Formula

Word limit $\approx$ `max_tokens` $\times$ **0.75**

Example

Desired output: ~100 words

Recommended setting: **`max_tokens` $\approx$ 130**

Pro Tip — Highest Reliability Technique

Use this exact instruction pattern:

Write exactly **50 words**. Count carefully.

Why This Works

It activates learned training patterns related to:

- Counting behavior
- Precision compliance

This significantly improves adherence.

Quick Summary

- Word limits can be controlled using clear instructions.
- This is a **soft behavioral constraint**, not a hard system limit.
- Tokens are the only strict control mechanism.
- Combining word limits with structured formats provides the best reliability.